

Validation Reliability, Not Model Capacity, Is the Bottleneck:

A Shift-Aware Multi-Source Ensemble for Sleep-Metric Prediction from Consumer Lifelog Sensors

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Abstract—We study the prediction of seven binary nightly sleep metrics from a single preceding day of consumer smartphone and smartwatch sensor logs, as posed by the ETRI 2026 Human Understanding AI Challenge. The task couples two difficulties that are common in real lifelog settings but rarely addressed together: a very small sample ($N=450$ training nights over only ten subjects) and a *temporal distribution shift*, where the evaluation nights fall predominantly later in time than the training nights. We build a nine-source ensemble spanning gradient-boosted trees, sequence neural networks, a frozen vision foundation model, an actigraphy-style sleep reconstruction, and a label-autocorrelation neighbour model, combined by per-target greedy blending. Our central finding is methodological: under temporal shift and small N , the binding constraint is not model capacity but *validation reliability*. Random k -fold cross-validation systematically rewards changes that fail to generalize, whereas only two classes of improvement transfer to held-out data—multi-seed variance reduction and *forward-chaining* (*shift-aware*) *reweighting*. Our final submission attains an Average Log-Loss of 0.578; more importantly, controlled ablations show that disciplined, shift-aware validation—rather than added model capacity—is what produced the improvements that generalized. We further give an honest analysis of the resulting data ceiling, which is structural: four of the seven labels derive from a clinical sleep sensor that is absent from the inputs.

Index Terms—lifelog, sleep prediction, ensemble learning, distribution shift, time-series cross-validation, small-sample learning

I. INTRODUCTION

Consumer wearables and smartphones passively record rich behavioural and physiological signals, raising the prospect of inferring sleep quality from everyday devices rather than dedicated clinical hardware. The ETRI 2026 Human Understanding AI Challenge formalizes one such task: given a single day of twelve passively logged sensor streams, predict seven binary descriptors of *that night's* sleep. Three descriptors (Q1–Q3) are subjective survey outcomes—sleep quality, fatigue, and stress—each binarized *relative to the individual's own long-run average*. The remaining four (S1–S4) encode whether the night met National Sleep Foundation (NSF) guidelines on

total sleep time, sleep efficiency, sleep-onset latency, and wake after sleep onset [1].

Two properties make the task unusually treacherous. First, the data are *small*: 450 training nights drawn from only ten subjects, with 250 evaluation nights from the same individuals. Second, there is a *temporal shift*: the evaluation nights lie largely after the training window for each subject (Fig. 1). Together these mean that a model can easily memorize subject-specific idiosyncrasies, and—more dangerously—that a naive validation estimate computed by randomly shuffling nights will be *optimistically biased* relative to the genuinely forward-in-time evaluation set.

The scoring metric is the macro (label-averaged) logarithmic loss,

$$\mathcal{L} = \frac{1}{7} \sum_{j=1}^7 -\frac{1}{N} \sum_{i=1}^N \left[y_{ij} \log \hat{p}_{ij} + (1 - y_{ij}) \log(1 - \hat{p}_{ij}) \right], \quad (1)$$

which rewards *calibrated probabilities* and punishes confident mistakes far more than a plain accuracy score would. Lower is better.

Our contributions are threefold:

- A diverse nine-source ensemble for consumer-sensor sleep-metric prediction, including an actigraphy-style sleep reconstruction that directly targets the NSF labels and a frozen vision-model embedding that injects an orthogonal view of the sensor time series (Section III).
- The empirical observation, made the centrepiece of the paper, that under temporal shift and small N the operative bottleneck is *validation reliability*: random-fold cross-validation misleads, and only variance reduction and forward-chaining reweighting transfer to held-out data (Sections IV–V).
- An honest characterization of the structural data ceiling that bounds achievable performance on this task, rooted in a clinical modality absent from the inputs (Section V).

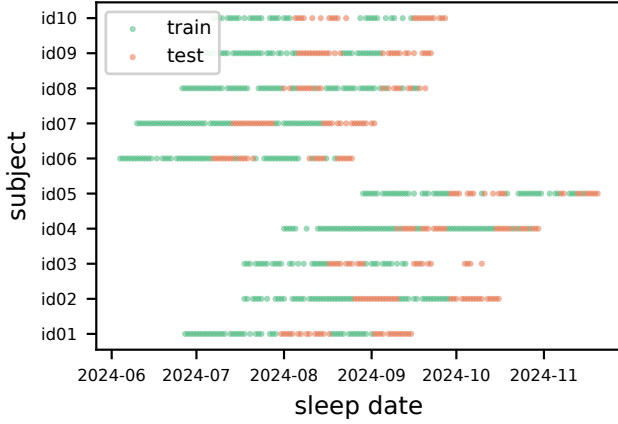


Fig. 1. Per-subject temporal distribution of training (green) and evaluation (orange) nights. Evaluation nights fall consistently later in time, defining the distribution shift the model must survive. This single observation motivates the forward-chaining validation of Section III.

II. DATA AND PROBLEM SETTING

Each night is associated with a `lifelog_date` (the preceding day whose sensors are observed) and a `sleep_date`. Twelve sensor families are provided: heart rate and pedometer from the watch; screen, charging, activity-state, ambient-sound, GPS, light, Wi-Fi, BLE, and app-usage from the phone. Sampling and coverage vary widely; the watch heart-rate stream in particular stores per-timestamp *arrays* of sub-second readings and is the sparsest at night.

a) Label structure: The two label families demand different inductive biases. Because Q1–Q3 are defined relative to a personal mean, their positive rate varies strongly across subjects (ranging roughly from 0.15 to 0.85 depending on subject and label); this motivates *within-subject normalization* of features and a ranking-based treatment of the Q targets. By contrast S1–S4 use *absolute* clinical thresholds, so a subject’s compliance rate is comparatively stable, and a per-subject base-rate prior is a strong anchor.

b) The missing modality: Critically, S1–S4 are computed from a Withings Sleep Analyzer—a bedside device that is *not* among the twelve input streams. The phone and watch can only *proxy* the quantities (total sleep time, efficiency, latency, wake) that those labels threshold. This gap is the source of the structural ceiling we report in Section V.

c) Temporal shift: Fig. 1 plots `sleep_date` for the training and evaluation nights per subject. For every subject the evaluation nights (orange) extend beyond the training nights (green). A validation protocol that shuffles nights uniformly therefore evaluates on a distribution the deployed model will never see; we return to this in Section III.

III. METHOD

Our system follows a three-stage structure: (i) build several diverse base sources, (ii) combine them per target by greedy

blending, and (iii) apply shift- and definition-aware corrections. All features are normalized within subject so that they encode “today relative to this person’s habit,” matching the relative definition of the Q labels.

A. Feature engineering

Each sensor is aggregated to one row per (subject, `lifelog_date`) over named hour windows (e.g. daytime, evening, pre-bed, night, morning) with the usual order statistics, plus night windows attributed to the correct calendar night. On top of the daily table we add per-subject temporal derivatives—lagged differences, rolling and exponential means, and an expanding mean—to encode personal baselines. The watch heart-rate arrays are reduced to per-minute mean/min/max/std after removing physiologically implausible values.

B. Base sources

We construct nine sources chosen for *mechanistic diversity*:

- 1) **Tree.** A weighted average of LightGBM [2], CatBoost [3] and Extremely Randomized Trees, with ExtraTrees importances selecting the top 150 features and a pseudo-labeling pass (confidence threshold 0.85) that adds high-confidence evaluation rows to training.
- 2) **Prior.** The per-subject historical label mean—a simple but strong anchor, especially for the absolute S labels.
- 3) **Sequence networks (GRU, BiLSTM, TCN [4], Attention).** Each is a small 1-D CNN encoder followed by the respective sequence module, a subject embedding, and a shared seven-head output. Each architecture is trained with eight random seeds and *averaged* (we call this *DLens*); since neural networks are high-variance learners on $N=450$, seed averaging reduces variance in the spirit of weight averaging [5].
- 4) **Vision embedding (SigLIP-Large).** The sensor series are rendered as 2-D images and passed through a *frozen* SigLIP-Large vision transformer [6]; the pooled embeddings feed a gradient-boosted model. The weights are never trained—the model supplies a generic visual encoding that is orthogonal to the tree and sequence views.
- 5) **Sleep reconstruction (SleepTree2).** An actigraphy-style routine reconstructs the main sleep episode from nocturnal movement, heart rate, and charging, yielding minute-level estimates of total sleep time, efficiency, latency and wake plus NSF-threshold flags; these feed a tree model and directly target S1–S4.
- 6) **Neighbour.** An exponentially date-weighted average of the same subject’s *other* nights’ labels (computed leave-one-out), exploiting the autocorrelation of sleep outcomes across nearby dates.

We additionally compute a label-free *day-state* representation by a transductive non-negative matrix factorization [9] of 30-minute behaviour bins over all 700 days, used to route the Q1/Q2 predictions.

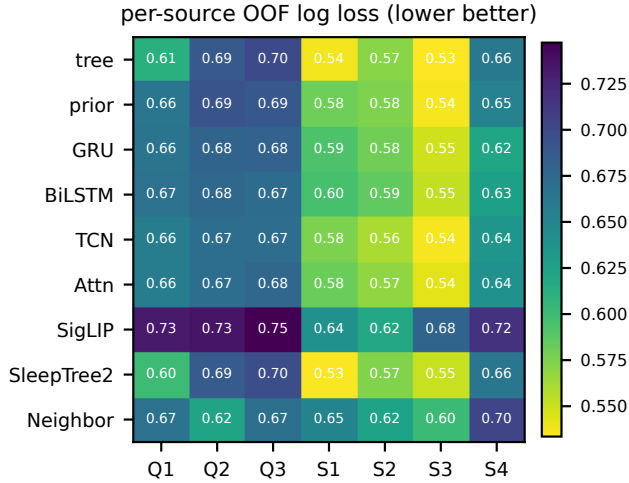


Fig. 2. Out-of-fold log loss of each base source per label (lighter is better). No source dominates every target, justifying the per-target blend.

C. Blending

For each target we select source weights by greedy coordinate descent, starting from the prior anchor and adjusting weights in $\pm 0.05/0.10$ steps to minimize (1). Fig. 2 shows that no single source dominates all labels—the sleep reconstruction is strongest on several S targets, the neighbour model on a Q target—which is precisely the condition under which blending pays.

D. Definition- and shift-aware corrections

Two corrections embody the paper’s thesis that the right inductive bias matters more than raw capacity.

a) *Pairwise ranking for Q*: Because the Q labels are defined by comparison to a personal average, we additionally learn a *pairwise* ranker [7]: for each subject we form (positive-night, negative-night) pairs and fit a ridge logistic model on their feature differences, adding the result as a scaled residual ($\alpha=0.15$ in logit space) to Q1/Q2.

b) *Forward-chaining reweighting for S*: This is the key lever. The blend weights above are chosen on randomly shuffled folds, which—as Fig. 1 warns—do not reflect the forward-in-time evaluation set. We therefore re-select the S1/S4 weights using *forward-chaining* folds: using the earliest 0.5 to 0.8 of nights for training, each fold validates on the following 20% block, sliding the split forward [8]. Fig. 3 shows that the forward protocol moves weight away from sources that look good only in-sample and toward sources that are robust to the shift. As we report next, this is one of only two changes that transfer to held-out performance.

IV. EXPERIMENTS

We report the macro log loss (1); lower is better, with a chance-level predictor at 0.693. We use internal out-of-fold (OOF) cross-validation during development and the challenge’s held-out test leaderboard for final evaluation. Our nine-

TABLE I
IMPROVEMENTS THAT TRANSFER TO HELD-OUT DATA ARE EXACTLY THOSE THAT REDUCE VARIANCE OR RESPECT THE TEMPORAL SHIFT; CHANGES THAT MERELY IMPROVE A RANDOM-FOLD ESTIMATE DO NOT TRANSFER.

Change	Random-fold OOF	Held-out
Multi-seed averaging (DLens)	improves	improves (small)
Forward-chaining S1/S4 reweighting	–	improves (largest)
High-divergence “new-signal” edits	improves	worsens

source backbone scores 0.584 in OOF; adding the definition- and shift-aware corrections of Section III yields a final held-out *Average Log-Loss* of 0.578. Because an absolute score is only meaningful against this particular benchmark, our analysis emphasizes *relative* comparisons, which are far more informative about what actually generalizes.

a) *What transferred*: Table I summarizes the two improvements that moved the held-out score and contrasts them with a representative “illusory” change. Multi-seed DLens averaging gave a small but reliable gain. The forward-chaining reweighting of S1/S4 was the largest single lever; held-out performance improved monotonically as we strengthened its influence. In contrast, candidates that diverged substantially from the current best prediction—even when they improved the random-fold OOF estimate—consistently *worsened* the held-out score, a pattern we observed repeatedly across roughly twenty tested ideas.

b) *Source diversity*: Diversity is what makes the blend pay. While the four sequence networks are, as expected, mutually correlated (up to 0.98), the mechanistically distinct families—trees, sleep reconstruction, neighbour, and the frozen vision embedding—are decorrelated to as low as 0.24 in their predictions. It is this *cross-family* disagreement, rather than within-family redundancy, that the per-target blend exploits, and it explains why the blend outperforms every individual source despite each source being weaker on its own.

V. DISCUSSION

a) *Why random CV misleads here*: The combination of small N , only ten subjects, and a forward-in-time evaluation set means a randomly shuffled fold leaks information in two ways: nearby nights of the same subject are highly autocorrelated, and the fold never reproduces the temporal gap the model faces at test time. Consequently the random-fold estimate rewards subtle overfitting. Forward-chaining validation removes the second leak and tracks held-out behaviour far better; it is the discipline, not a new feature, that produced our gains.

b) *The structural ceiling*: Despite extensive search—vision models, multiple sleep-reconstruction variants, heart-rate-variability textures, per-subject models, external weather and connectivity features, and several label-structure exploits—no “big bet” produced a held-out gain once validated without leakage. The reason is structural: S1–S4 threshold quantities measured by a clinical device absent

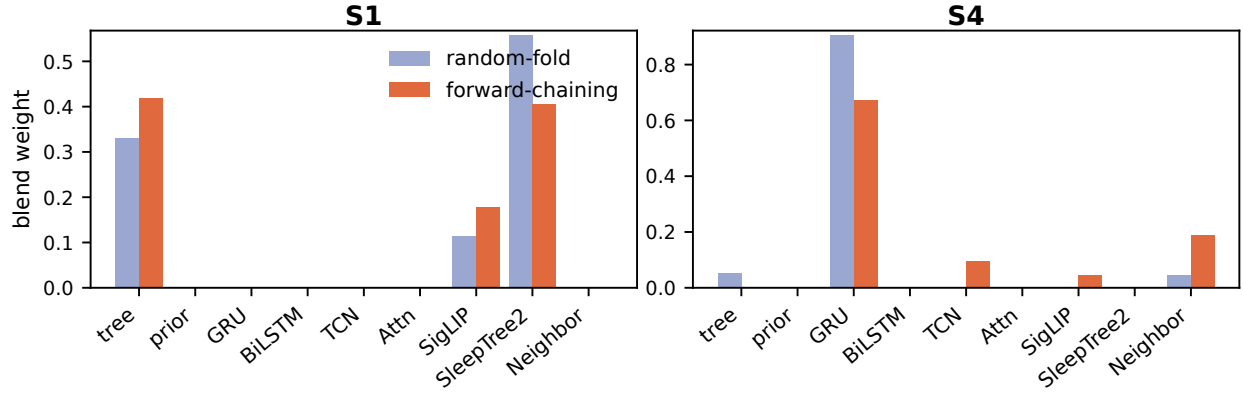


Fig. 3. Blend weights for S1 and S4 chosen by random-fold (blue) versus forward-chaining (orange) validation over the same nine sources. The shift-aware protocol redistributes weight toward sources that generalize forward in time (e.g., on S4 away from the in-sample-favoured GRU toward TCN and the neighbour model); this reweighting is the single largest source of held-out improvement.

from the inputs, and Q2–Q3 are subjective states only loosely determined by behaviour. The phone and watch can proxy these only so far, which bounds the achievable score. We regard the honest demarcation of this ceiling as a useful negative result for the lifelog community.

VI. CONCLUSION

For nightly sleep-metric prediction from consumer lifelog sensors under small samples and temporal shift, we found that the decisive factor was not model capacity but the reliability of the validation protocol. A mechanistically diverse nine-source ensemble provided a strong backbone, but the improvements that actually generalized were those that reduced variance (seed averaging) or respected the temporal shift (forward-chaining reweighting); changes that merely improved a random-fold estimate did not transfer. We additionally documented a structural data ceiling rooted in a missing clinical modality. We hope these observations help practitioners working with similarly small, shifted lifelog datasets prioritize honest validation over added capacity.

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